



EASYVISA - CASE STUDY

Project Report

Project Details:

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1. Project Background

Business communities in the United States are experiencing a high demand for human resources, and one of their ongoing challenges is finding and attracting the right talent, which is crucial for maintaining competitiveness. Companies in the US seek hardworking, skilled, and qualified individuals both domestically and internationally.

The Immigration and Nationality Act (INA) allows foreign workers to enter the United States for temporary or permanent employment. This act also protects US workers from any negative impacts on their wages or working conditions by requiring employers to adhere to statutory requirements when hiring foreign workers to address labour shortages. The Office of Foreign Labor Certification (OFLC) manages these immigration programs.

OFLC reviews job certification applications from employers who wish to bring foreign workers to the US. It grants certifications if employers can prove there are not enough US workers available to perform the job at wages that meet or exceed the standard for the occupation in the intended employment area.

2. Objective

In FY 2016, the Office of Foreign Labor Certification (OFLC) processed 775,979 employer applications for 1,699,957 positions requiring temporary or permanent labour certifications. This represented a 9% increase in the number of applications processed compared to the previous year. As the volume of applications grows annually, reviewing each case is becoming increasingly labour-intensive.

To address this challenge, there is a need for a Machine Learning-based solution to assist in shortlisting candidates with a higher likelihood of visa approval. The OFLC has engaged EasyVisa to provide data-driven solutions.

Our primary focus is to analyse the data using classification model to:

- Streamline the visa approval process.
- Recommend whether to certify or deny visas for applicants, based on key factors that significantly impact the case status.

3. Data description

The data contains different attributes of employee application details. The detailed data dictionary is given below.

- **case_id:** ID of each visa application
- **continent:** Information of continent the employee
- **education_of_employee:** Information of education of the employee
- **has_job_experience:** Does the employee has any job experience? Y= Yes; N = No
- **requires_job_training:** Does the employee require any job training? Y = Yes; N = No
- **no_of_employees:** Number of employees in the employer's company
- **yr_of_estab:** Year in which the employer's company was established
- **region_of_employment:** Information of foreign worker's intended region of employment in the US.
- **prevailing_wage:** Average wage paid to similarly employed workers in a specific occupation in the area of intended employment. The purpose of the prevailing wage is to ensure that the foreign worker is not underpaid compared to other workers offering the same or similar service in the same area of employment.
- **unit_of_wage:** Unit of prevailing wage. Values include Hourly, Weekly, Monthly, and Yearly.
- **full_time_position:** Is the position of work full-time? Y = Full-Time Position; N = Part-Time Position
- **case_status:** Flag indicating if the Visa was certified or denied.

4. Exploratory Data Analysis

4.1. Data structure

After importing NumPy and Pandas libraries for tasks like data reading, data computation and data manipulation and loading data, the following observations regarding the data structure were noted.

	case_id	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees
0	EZYV01	Asia	High School	N	N	14513
1	EZYV02	Asia	Master's	Y	N	2412
2	EZYV03	Asia	Bachelor's	N	Y	44444
3	EZYV04	Asia	Bachelor's	N	N	98
4	EZYV05	Africa	Master's	Y	N	1082

yr_of_estab	region_of_employment	prevailing_wage	unit_of_wage	full_time_position	case_status
2007	West	592.2029	Hour	Y	Denied
2002	Northeast	83425.6500	Year	Y	Certified
2008	West	122996.8600	Year	Y	Denied
1897	West	83434.0300	Year	Y	Denied
2005	South	149907.3900	Year	Y	Certified

Table 1. Data structure

OBSERVATION	VARIABLES
25480	12

4.2. Data types

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25480 entries, 0 to 25479
Data columns (total 12 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                ---
0   case_id                              25480 non-null  object
1   continent                            25480 non-null  object
2   education_of_employee                25480 non-null  object
3   has_job_experience                   25480 non-null  object
4   requires_job_training                25480 non-null  object
5   no_of_employees                     25480 non-null  int64
6   yr_of_estab                         25480 non-null  int64
7   region_of_employment                25480 non-null  object
8   prevailing_wage                     25480 non-null  float64
9   unit_of_wage                        25480 non-null  object
10  full_time_position                  25480 non-null  object
11  case_status                         25480 non-null  object
dtypes: float64(1), int64(2), object(9)
memory usage: 2.3+ MB
```

Table 2. Data types

- The data set contains twelve variables indexed from 0-11.
- The data set consists of 3 numerical columns, where 2 are of integer data type and 1 is float data type and 9 object columns.
- The case_id column does not add value to the data. Hence can be dropped.
- The case_status column is the target variable.

4.3. Null values

- No null values were found in the data set.

4.4. Duplicate Values

- No duplicate values were found in the data set.

4.5. Statistical Summary

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
continent	25480	6	Asia	16861	NaN	NaN	NaN	NaN	NaN	NaN	NaN
education_of_employee	25480	4	Bachelor's	10234	NaN	NaN	NaN	NaN	NaN	NaN	NaN
has_job_experience	25480	2	Y	14802	NaN	NaN	NaN	NaN	NaN	NaN	NaN
requires_job_training	25480	2	N	22525	NaN	NaN	NaN	NaN	NaN	NaN	NaN
no_of_employees	25480.0	NaN	NaN	NaN	5667.04321	22877.928848	-26.0	1022.0	2109.0	3504.0	602069.0
yr_of_estab	25480.0	NaN	NaN	NaN	1979.409929	42.366929	1800.0	1976.0	1997.0	2005.0	2016.0
region_of_employment	25480	5	Northeast	7195	NaN	NaN	NaN	NaN	NaN	NaN	NaN
prevailing_wage	25480.0	NaN	NaN	NaN	74455.814592	52815.942327	2.1367	34015.48	70308.21	107735.5125	319210.27
unit_of_wage	25480	4	Year	22962	NaN	NaN	NaN	NaN	NaN	NaN	NaN
full_time_position	25480	2	Y	22773	NaN	NaN	NaN	NaN	NaN	NaN	NaN
case_status	25480	2	Certified	17018	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Table 3. Statistical summary

The following observations were made:

- **continent:** The applications come from all six continents, with Asia leading the list with a frequency of 16861.
- **education_of_employee:** The majority of applicants hold a bachelor's degree.
- **has_job_experience:** 58% of the applicants have work experience.
- **requires_job_training:** 88% of the applicants do not require job training, indicating that most have relevant experience.
- **no_of_employees:** On average, companies employ approximately 5,667 people. Negative values are found in the column which indicates an error and has to be treated.
- **yr_of_estab:** The establishment years of the employer's company range from 1800 to 2016.
- **region_of_employment:** The intended region of employment for foreign workers in the US is predominantly the Northeast, with a frequency of 7,195.
- **prevailing_wage:** The average wage for similarly employed workers in a specific occupation within the intended employment area ranges from \$2.13 to \$319,210, with an average of \$74,455. Outliers are seen in the data.

- **unit_of_wage:** Among hourly, weekly, monthly, and yearly wages, yearly wages are the most prevalent, accounting for approximately 90%.
- **full_time_position:** Approximately 89% of the available job positions are full-time.
- **case_status:** 66.7% of the visa applications are approved.

4.6. Count and Percentage of Categorical Variable in the Data Set

continent Asia 16861 Europe 3732 North America 3292 South America 852 Africa 551 Oceania 192 Name: count, dtype: int64	continent Asia 0.661735 Europe 0.146468 North America 0.129199 South America 0.033438 Africa 0.021625 Oceania 0.007535 Name: proportion, dtype: float64
education_of_employee Bachelor's 10234 Master's 9634 High School 3420 Doctorate 2192 Name: count, dtype: int64	education_of_employee Bachelor's 0.401648 Master's 0.378100 High School 0.134223 Doctorate 0.086028 Name: proportion, dtype: float64
has_job_experience Y 14802 N 10678 Name: count, dtype: int64	has_job_experience Y 0.580926 N 0.419074 Name: proportion, dtype: float64
requires_job_training N 22525 Y 2955 Name: count, dtype: int64	requires_job_training N 0.884027 Y 0.115973 Name: proportion, dtype: float64
region_of_employment Northeast 7195 South 7017 West 6586 Midwest 4307 Island 375 Name: count, dtype: int64	region_of_employment Northeast 0.282378 South 0.275392 West 0.258477 Midwest 0.169035 Island 0.014717 Name: proportion, dtype: float64
unit_of_wage Year 22962 Hour 2157 Week 272 Month 89 Name: count, dtype: int64	unit_of_wage Year 0.901177 Hour 0.084655 Week 0.010675 Month 0.003493 Name: proportion, dtype: float64
full_time_position Y 22773 N 2707 Name: count, dtype: int64	full_time_position Y 0.89376 N 0.10624 Name: proportion, dtype: float64
case_status Certified 17018 Denied 8462 Name: count, dtype: int64	case_status Certified 0.667896 Denied 0.332104 Name: proportion, dtype: float64

Table 4. Count and percentage of categorical variables

4.7. Univariate analysis

4.7.1. Number of employees

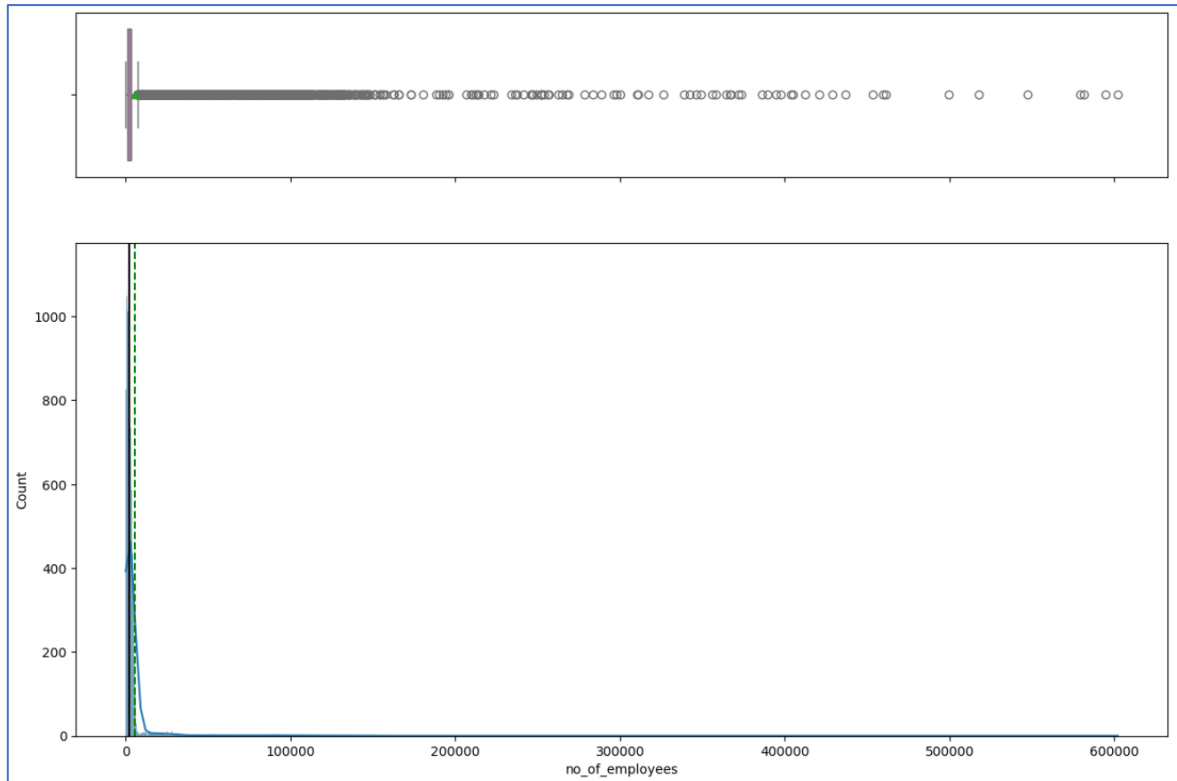


Fig 1. Number of employees

OBSERVATIONS:

- The number of employees employed by the company ranges from 12 to 600000.
- The mean and median are close to each other, indicating a normal distribution which is highly skewed towards the right.
- The count is high between 0 and 5,000, but decreases starting from 50,000 and remains steady up to 600,000.
- The range indicates that employees are required for a variety of job roles.
- Negative value is seen in the data, which might be an error that has to be treated.

4.7.2. Prevailing wages

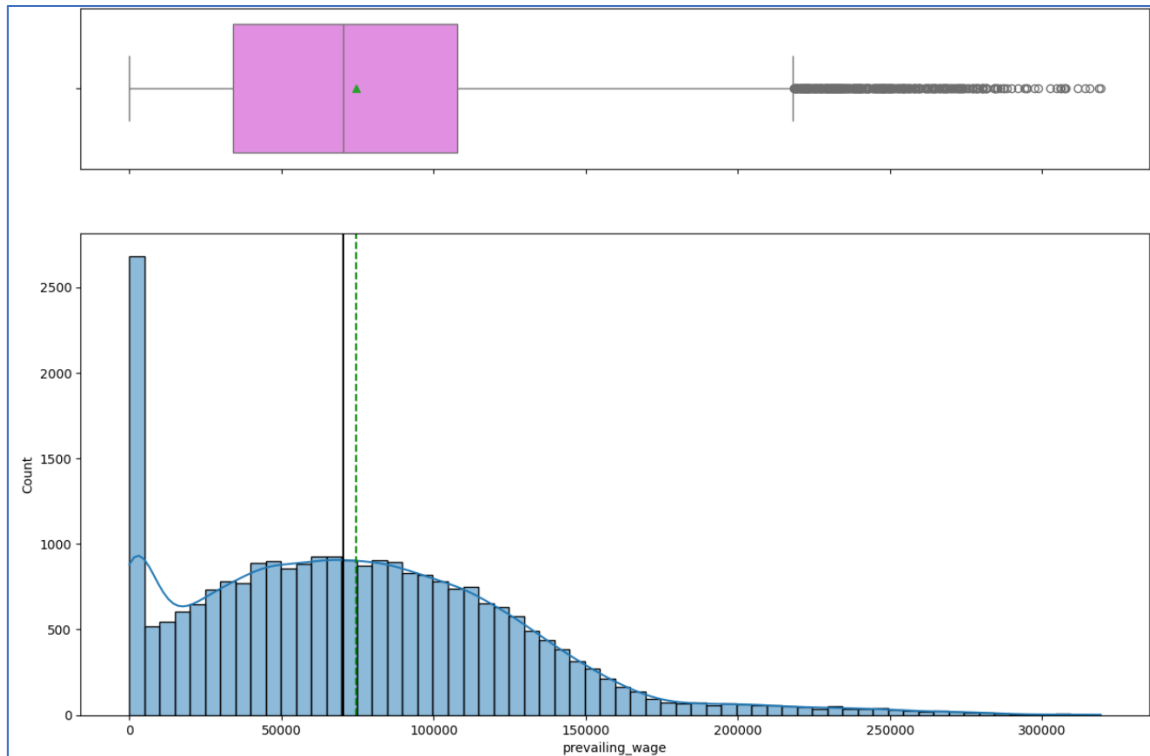


Fig 2. Prevailing wage

OBSERVATIONS:

- The prevailing wages range from \$2.13 to \$319,210.
- The frequency of prevailing wages is highest between \$2.13 and \$5,000.
- The distribution is nearly bimodal, with a rightward skew.
- The frequency of prevailing wages is low above \$200,000

4.7.3. Year of establishment

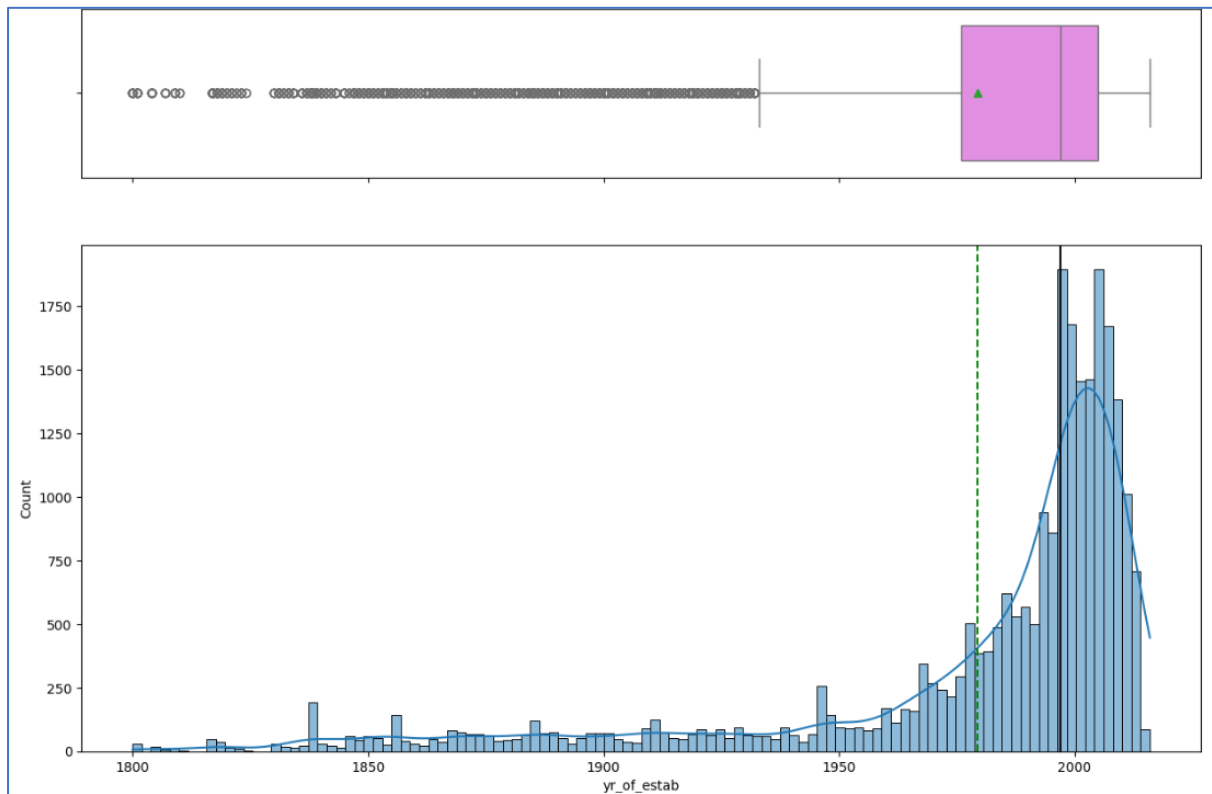


Fig 3. Year of establishment

OBSERVATIONS:

- The number of companies established in the late 1990's to early 2000's are higher.
- The distribution is a highly left skewed distribution.

4.7.4. Continent

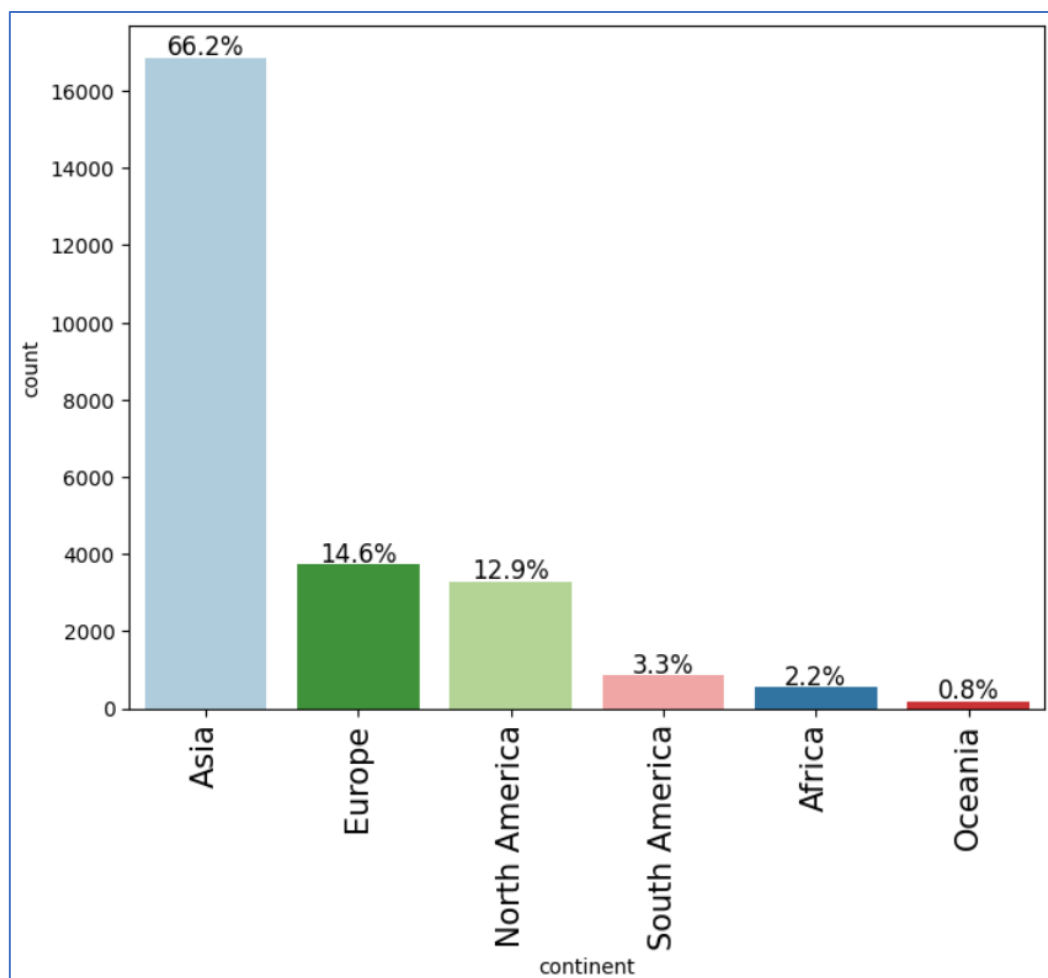


Fig 4. Continent

OBSERVATIONS:

- Applications come from all six continents, with Asia at the top with 66.2%, and Oceania at the bottom with 0.8%.
- Applications from Europe and North America account for 14.6% and 12.9%, respectively.

4.7.5. Education of employee

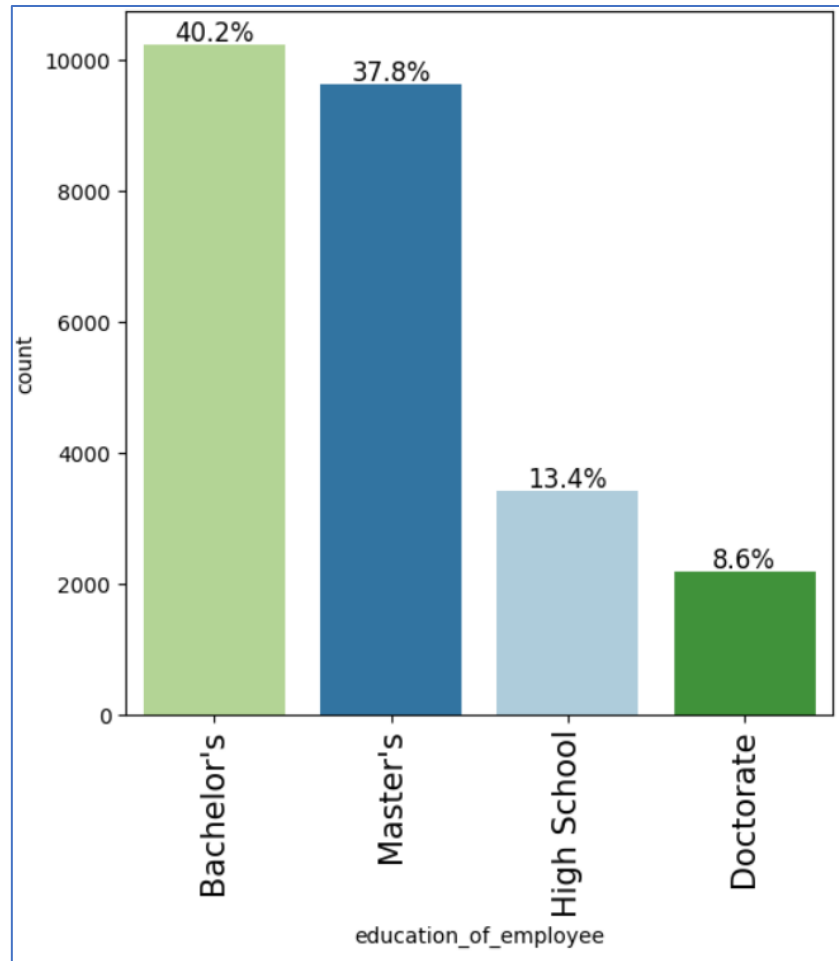


Fig 5. Education of employee

OBSERVATIONS:

- Most employees hold a bachelor's degree, with 40.2%, while those with a master's degree come in second at 37.8%.
- Employees with a doctorate rank the lowest, at 8.6%, suggesting that those with doctoral degrees are more likely to remain in jobs within their own countries rather than seeking opportunities abroad.

4.7.6. Has job experience

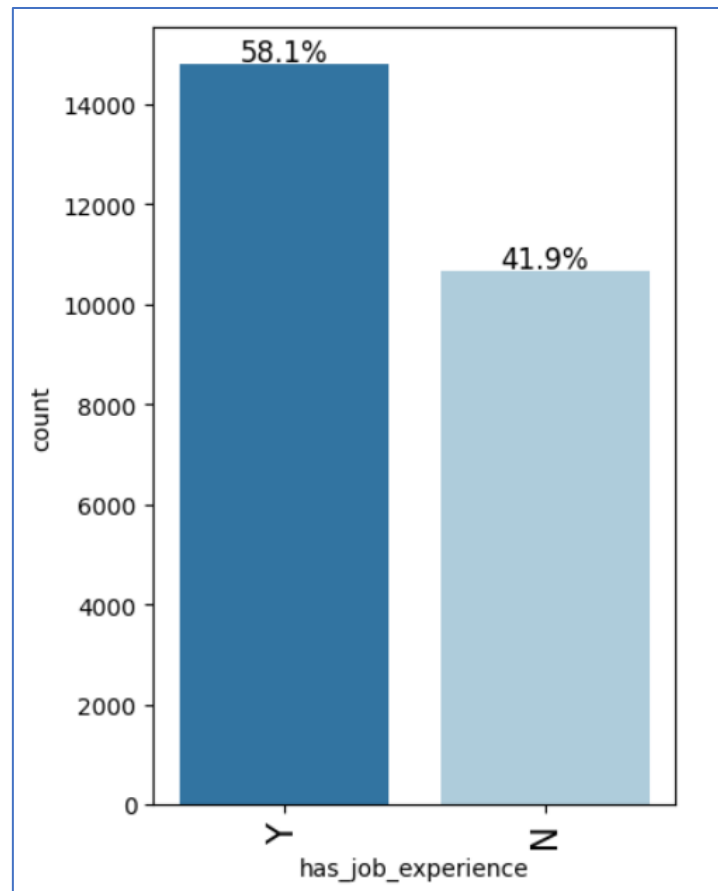


Fig 6. Has job experience

OBSERVATIONS:

- Most employees have prior job experience, with the percentage reaching 58.1%.
- This suggests that the majority of employed individuals are seeking positions that align with their relevant work experience.

4.7.7. Requires job training

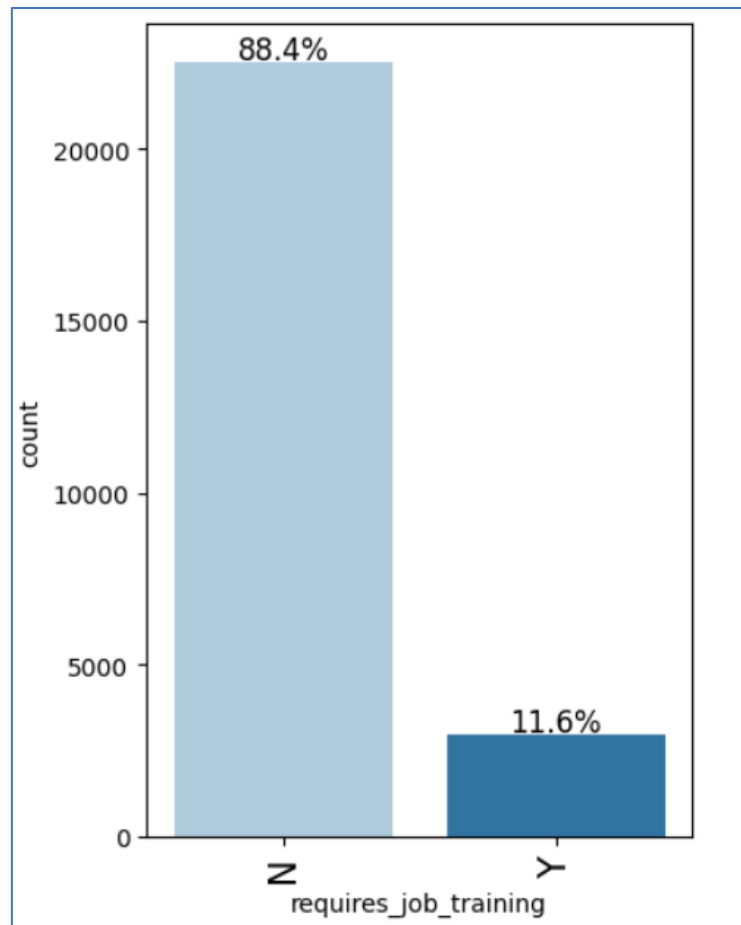


Fig 7. Requires job training

OBSERVATIONS:

- Most employees, accounting for 88.4%, do not require job training.
- This suggests that most employees are applying for positions that align with their qualifications and standards.

4.7.8. Region of employment

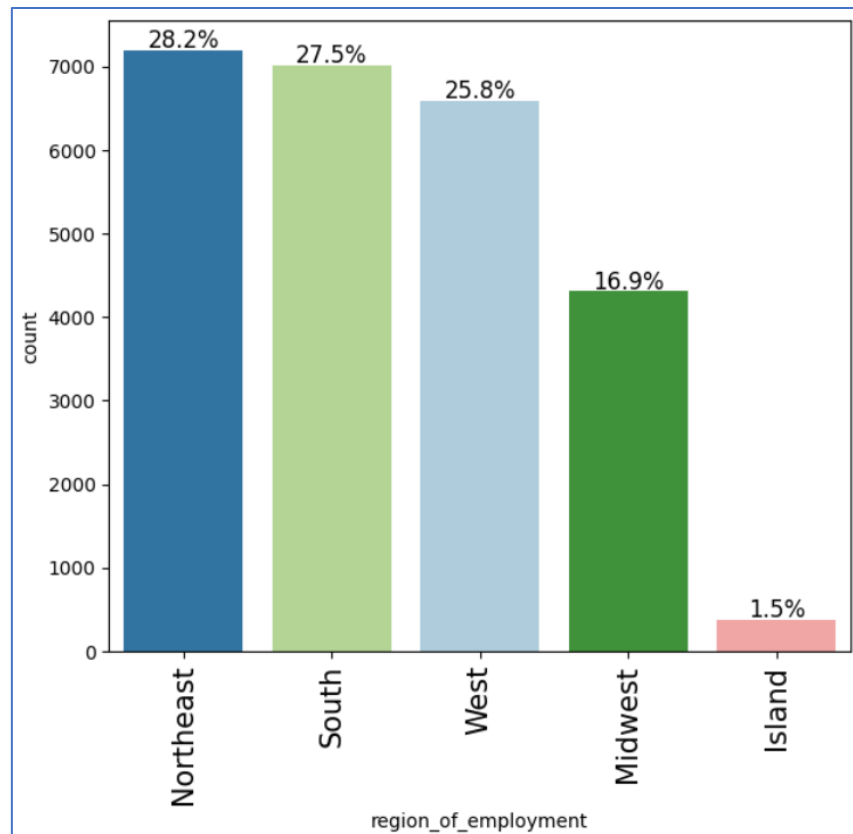


Fig 8. Region of employment

OBSERVATIONS:

- Most employment opportunities are found in the Northeast and South regions of the U.S., with percentages of 28.2% and 27.5%, respectively.
- The West of the U.S. also offers significant employment opportunities, accounting for 25.8%.
- Iceland has the lowest employment rate, at 1.5%.

4.7.9. Unit of wage

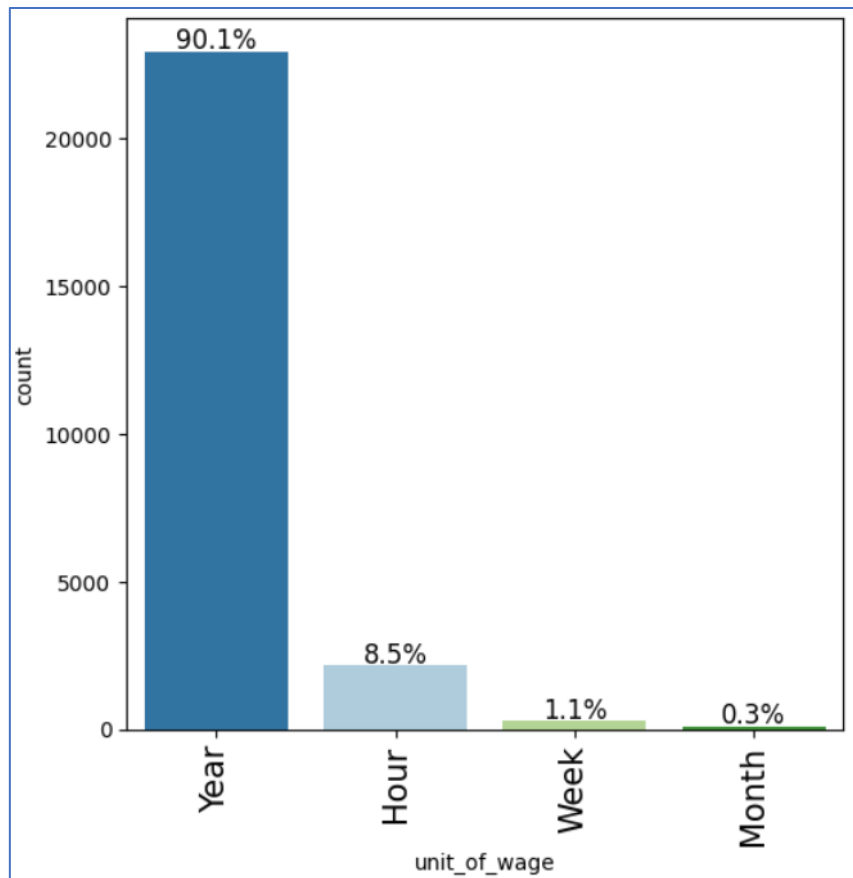


Fig 9. Unit of wage

OBSERVATIONS:

- In the target occupation in the area, 90.1% of similarly employed workers receive annual salaries.
- The percentage of prevailing wages paid on an hourly, weekly, or monthly basis is low.

4.7.10. Full time position

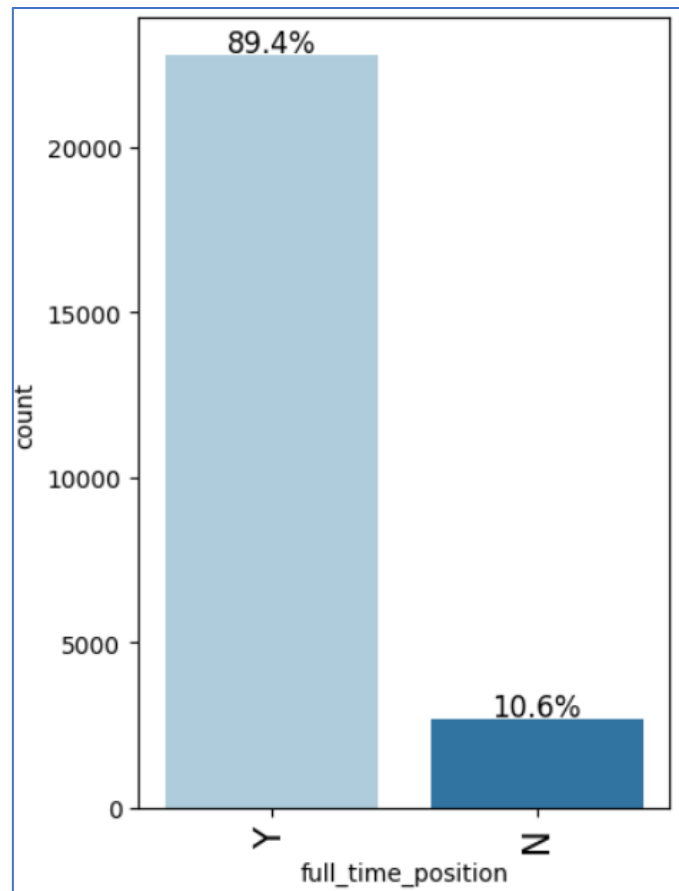


Fig 10. Full time position

OBSERVATIONS:

- The majority of employees (89.4%) have applied for full-time positions.
- The 10.6% of employees who have applied for part-time positions may be students.

4.7.11. Case status

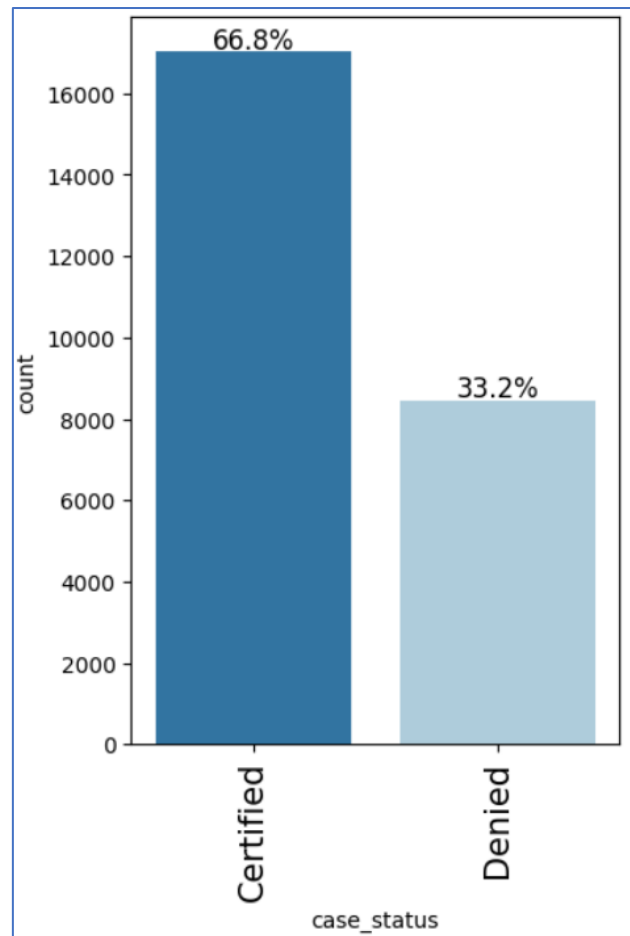


Fig 11. Case status

OBSERVATIONS:

- 66.8% of the visa applications have been approved.
- The number of rejected applications is also substantial.

4.8 Bivariate analysis

4.8.1 Heat Map

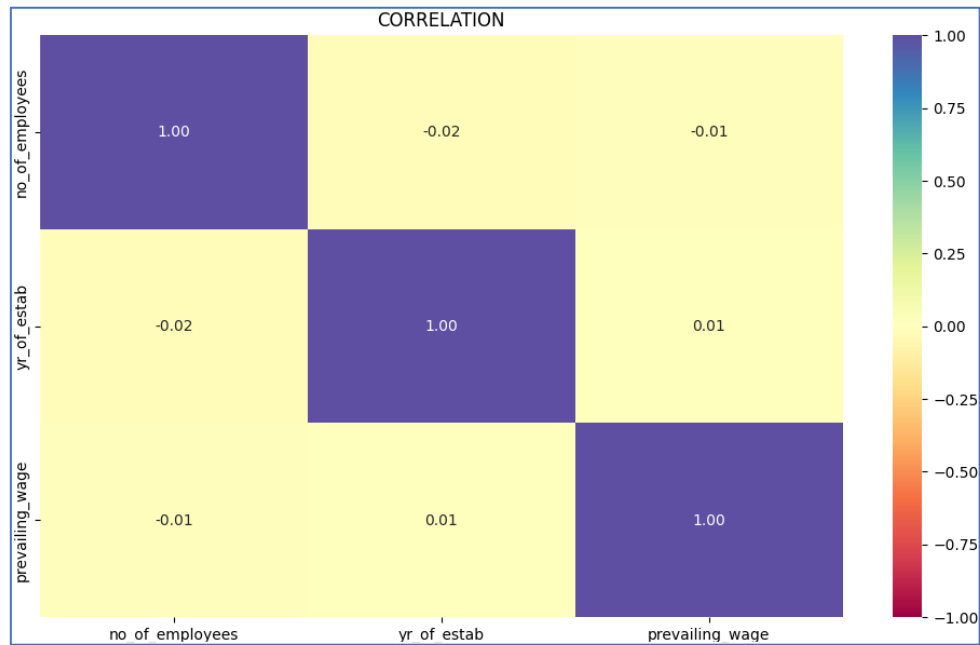


Fig 12. Heat map

OBSERVATIONS:

- The heatmap shows that there is no correlation between the numerical variables.
- The correlation between the numerical variables shows values as low as -0.01, -0.02 and 0.01.
- This suggests that the variables do not have a linear relationship with each other, but it does not rule out the possibility of more complex relationships or interactions. Further analysis may be required to fully understand the dynamics between the variables.

4.8.2 Pair Plot for understanding the relationship between numerical variables

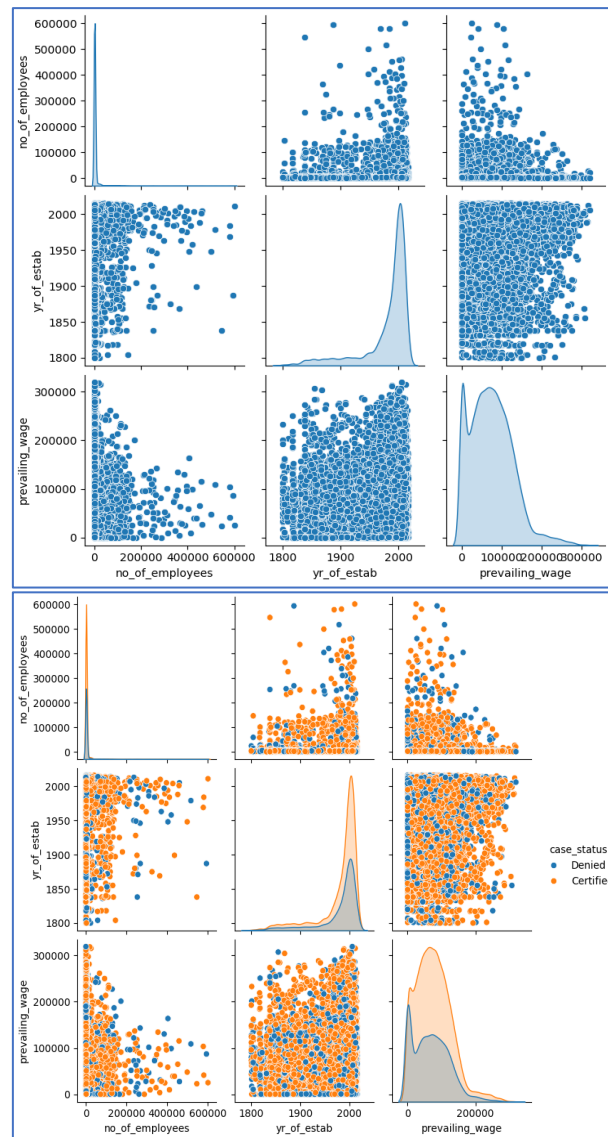


Fig 13. Pair plot

OBSERVATIONS:

- The number of employees tends to increase with the company's year of establishment. However, the density is higher among older companies.
- The number of employees is higher when the prevailing wage is low, indicating that there are comparatively more employees earning middle-level wages.
- The distribution of prevailing wage and year of establishment is almost uniformly distributed.
- The distribution for both visa approvals and denials appear nearly identical, with cases of denial being comparatively fewer.

4.8.3 Case status vs No of employees

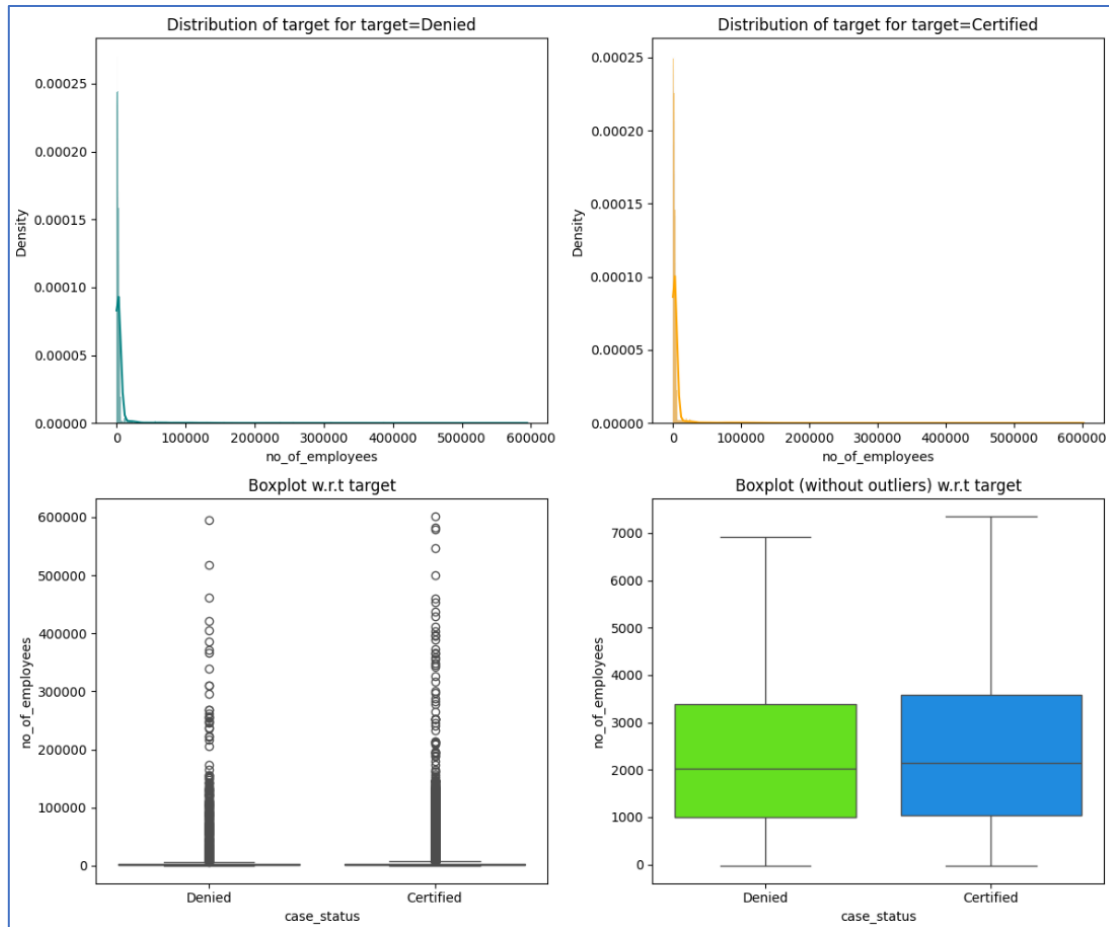


Fig 14. Case status vs No of employees

OBSERVATIONS:

- The number of employees in the employer company does not significantly affect whether a visa is approved or denied.

4.8.4 Case status vs Prevailing wage

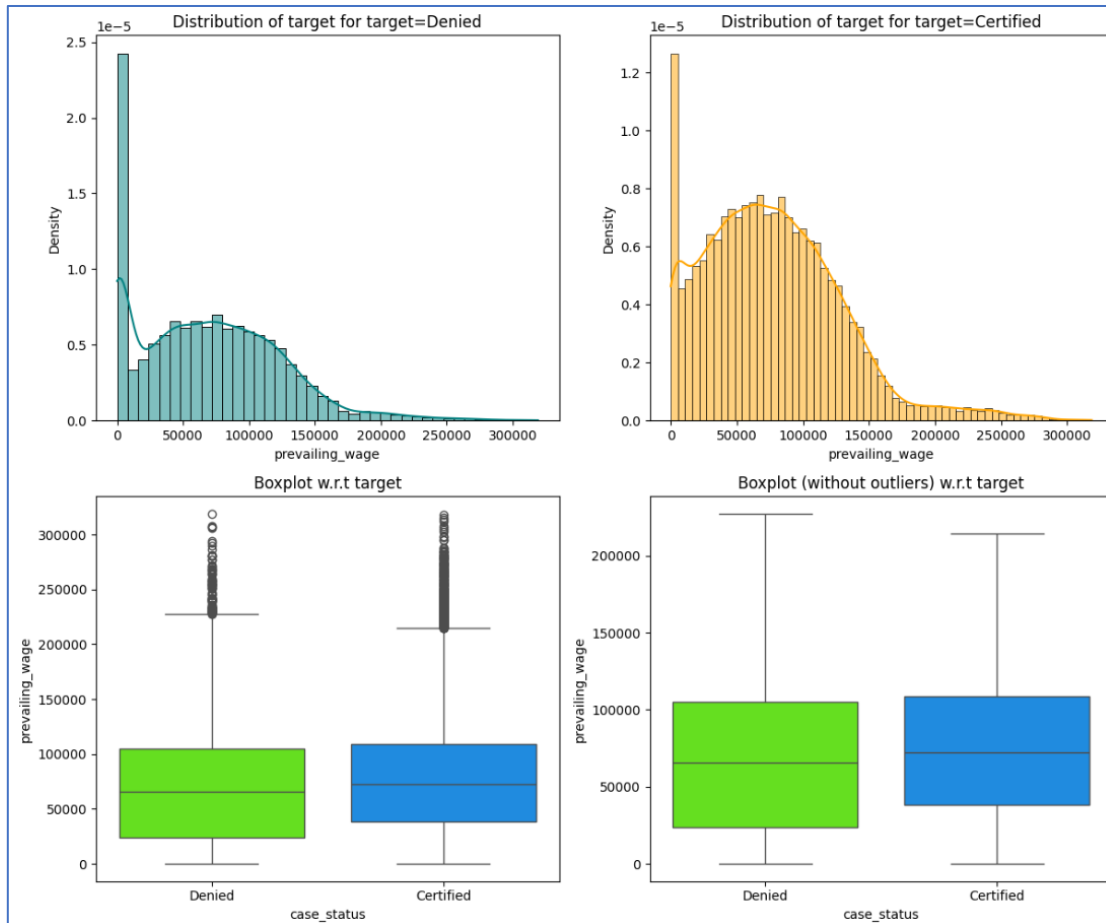


Fig 15. Case status vs Prevailing wage

OBSERVATIONS:

- The distribution pattern of the prevailing wage is similar for both certified and denied visa applications.
- The mean prevailing wage for both certified and denied visa applications is nearly identical.

4.8.5 Case status vs Year of establishment

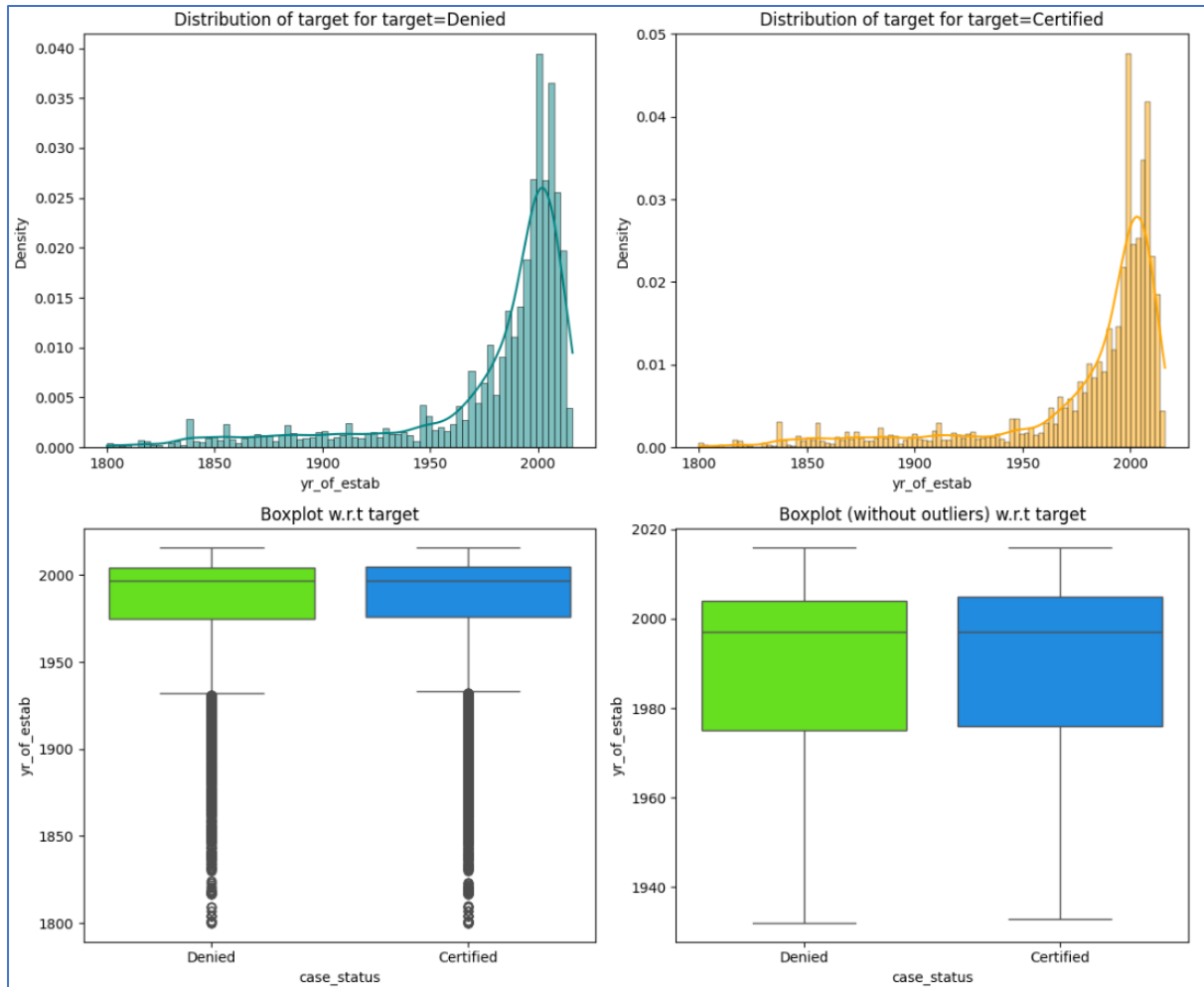


Fig 16. Case status vs Year of establishment

OBSERVATIONS:

- The year of establishment of the employer company does not significantly impact whether the visa is certified or denied.

4.8.6 Case status vs Continent

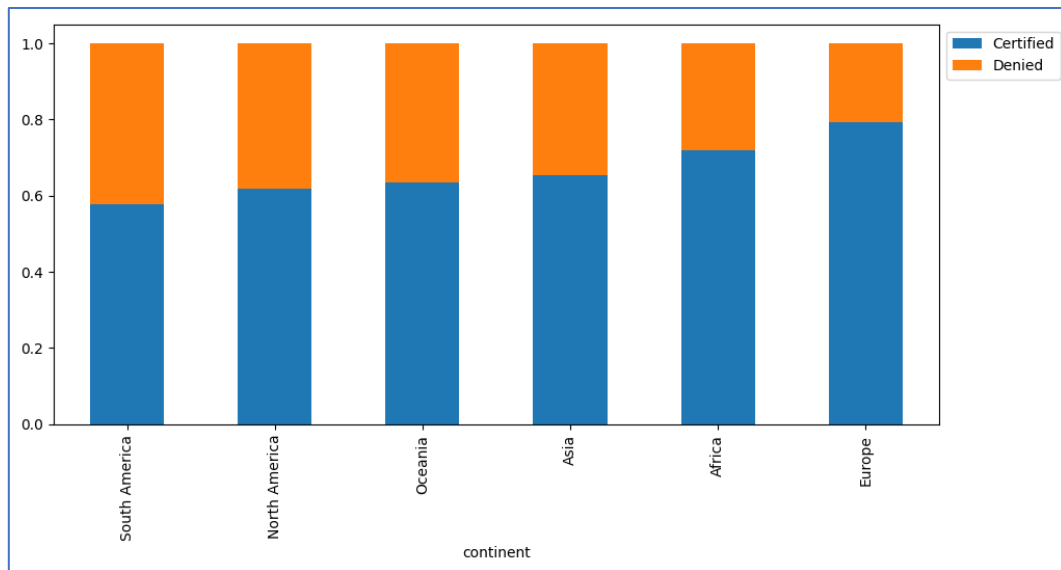


Fig 17. Case status vs Continent

OBSERVATIONS:

- Almost 65.3% of the certified visa applicants belong to Asia.

4.8.7 Case status vs Education of employee

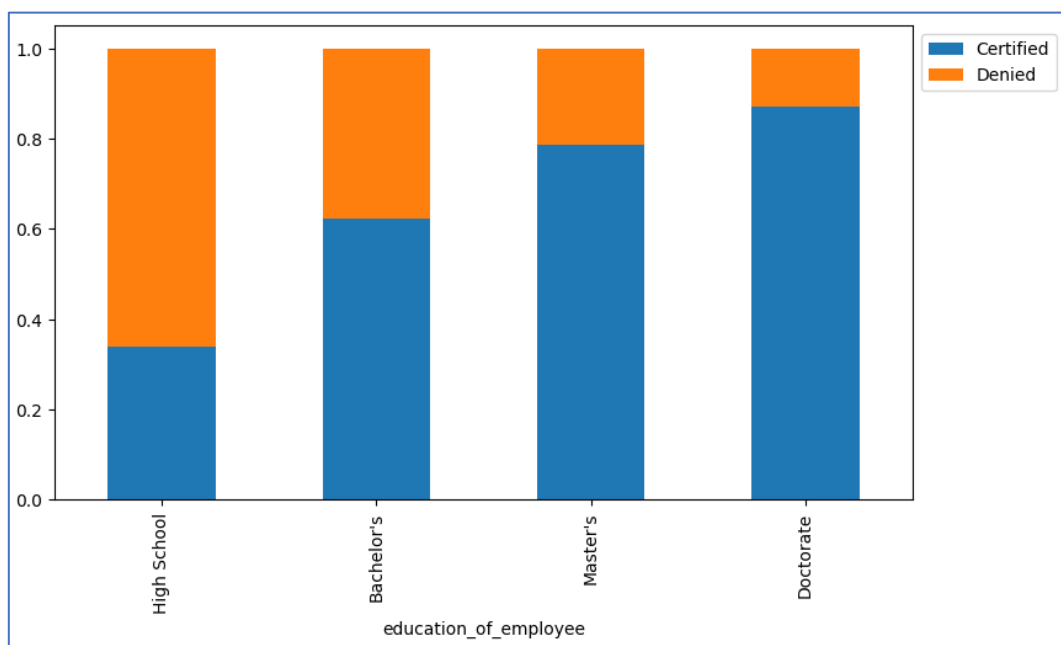


Fig 18. Case status vs Education of employee

OBSERVATIONS:

- 66.7% of visa applications are approved. Of these approvals, 25% are granted to employees with a bachelor's degree, who rank first, followed by those with a master's degree.

4.8.8 Case status vs Has job experience

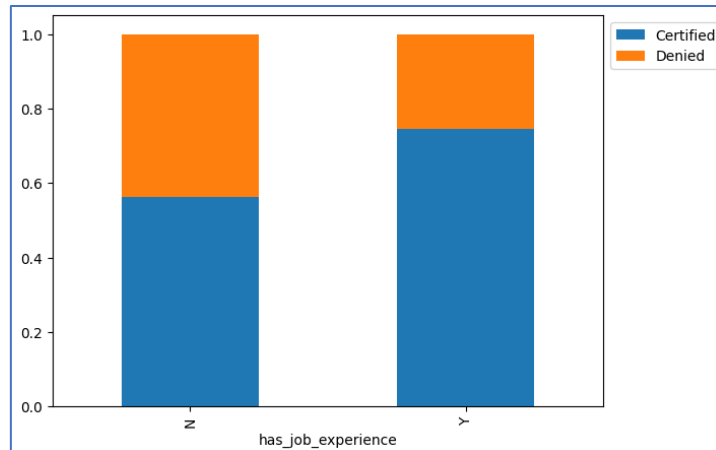


Fig 19. Case status vs Has job experience

OBSERVATIONS:

- 64.7% of employees with job experience are granted a visa.

4.8.9 Case status vs Requires job training

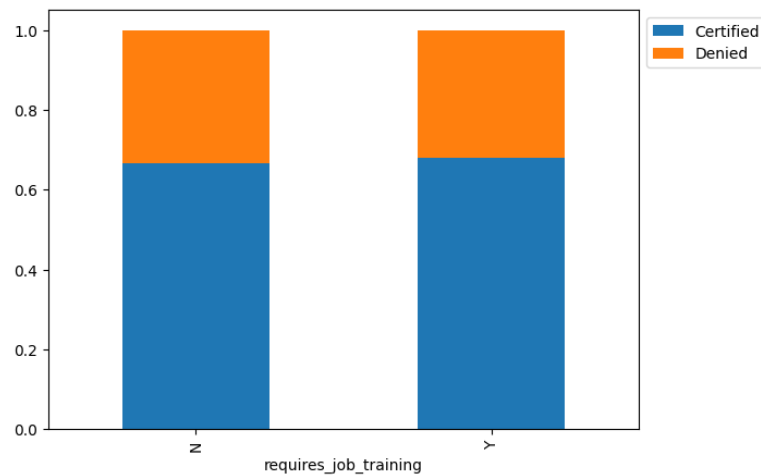


Fig 20. Case status vs Requires job training

OBSERVATIONS:

- 88.2% of employees who do not need job training are granted a visa, which is significantly higher than the percentage for those who do require job training.
- Additionally, the percentage of visa denials is also notably high for those who do not require job training.

4.8.10 Case status vs Region of employment

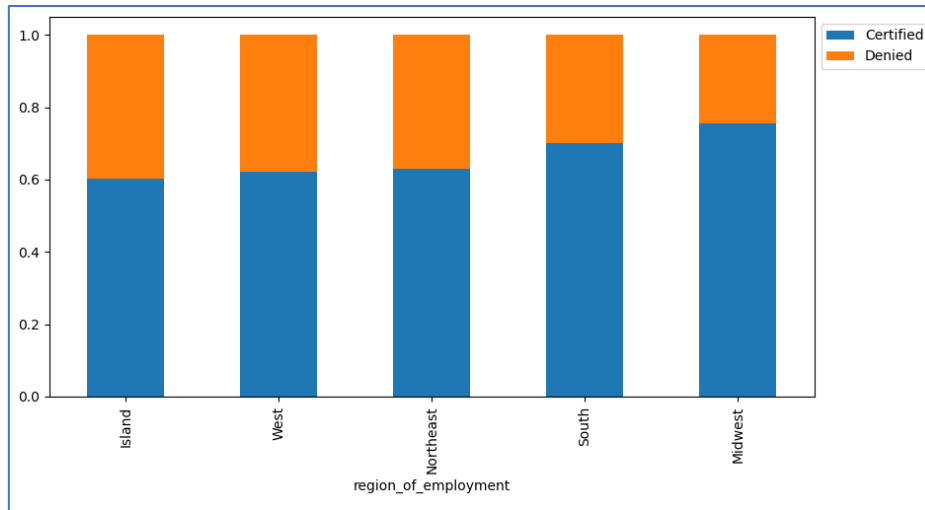


Fig 21. Case status vs Region of employment

OBSERVATIONS:

- Employment opportunities in the Northeast (26.5%) and West regions (25%) of the US have higher visa approval rates compared to the Island region, which has the lowest rates.

4.8.11 Case status vs Unit of wage

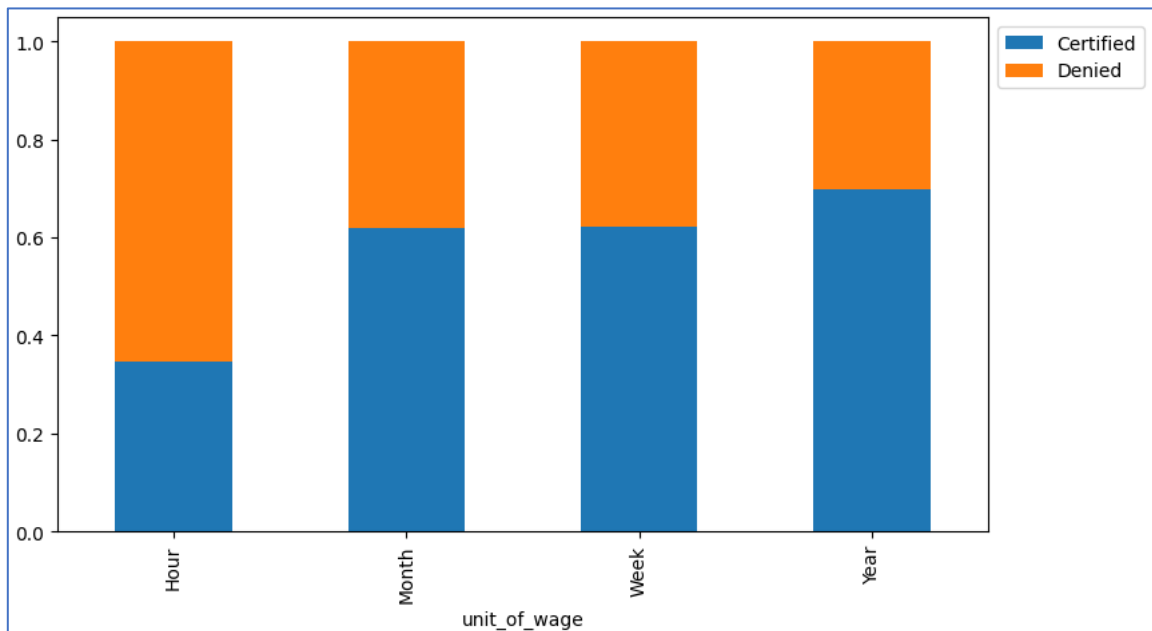


Fig 22. Case status vs Unit of wage

OBSERVATIONS:

- 81% of visa denials occur in employment opportunities with annual wage structures, while only 0.4% of visa denials are associated with monthly wages, which represents the lowest rate.

4.8.12 Case status vs Full time position

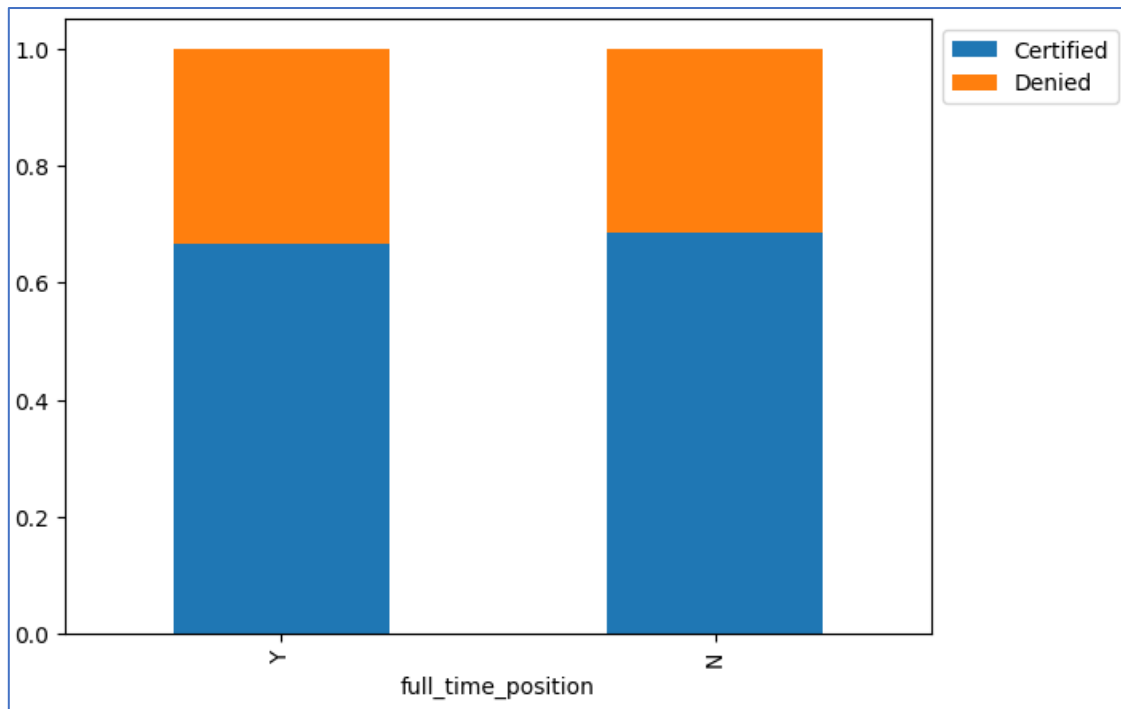


Fig 23. Case status vs Full time position

OBSERVATIONS:

- Both visa denials and approvals are high for full-time job requirements, but visa approvals have a higher rate.

4.9 Insights based on Exploratory Data Analysis

4.9.1 Based on Univariate analysis:

- **no_of_employees:** The number of employees at these companies varies from 12 to 600,000, with a significant rightward skew in the distribution. There is a high concentration of companies with employee counts between 0 and 5,000, but the frequency decreases starting from 50,000 and remains consistent up to 600,000. This range reflects the diverse range of job roles required across different companies.
- **prevailing_wage:** The prevailing wages span from \$2.13 to \$319,210, with the highest frequency observed in the range between \$2.13 and \$5,000.
- **yr_of_estab:** The number of companies established from the late 1990s to the early 2000s is notably higher, with the distribution showing a pronounced leftward skew.
- **continent:** Applications are received from all six continents, with Asia leading at 66.2% and Oceania at the lowest with 0.8%. Europe and North America contribute 14.6% and 12.9% of the applications, respectively.
- **education_of_employee:** The majority of employees hold a bachelor's degree, accounting for 40.2%, followed by those with a master's degree at 37.8%. Doctorate holders are the least represented at 8.6%, indicating that individuals with doctoral degrees are more likely to stay in their home countries rather than pursuing opportunities abroad.
- **has_job_experience:** Most employees have prior job experience, with 58.1% reporting such backgrounds. This indicates that the majority of employed individuals are pursuing positions that align with their previous work experience.
- **requires_job_training:** Most employees, comprising 88.4%, do not require job training. This suggests that the majority are applying for positions that match their existing qualifications and expertise.
- **region_of_employment:** The majority of employment opportunities are concentrated in the Northeast and South regions of the U.S., with 28.2% and 27.5% respectively. The West also provides substantial employment prospects, representing 25.8%. Iceland has the lowest employment rate, at just 1.5%.
- **unit_of_wage:** In the target occupation within the area, 90.1% of similarly employed workers receive annual salaries. The proportion of workers paid on an hourly, weekly, or monthly basis is relatively low.
- **full_time_position:** Most employees, about 89.4% have applied for full-time positions, while the 10.6% who have applied for part-time roles may be students.
- **case_status:** 66.8% of visa applications have been approved, though a significant number of applications have also been rejected.

4.9.2 Based on Bivariate analysis:

- The size of the employer company does not have a significant impact on whether a visa is approved or denied.
- The number of employees generally rises with the company's age, but the density is greater among older companies.
- When the prevailing wage is low, the number of employees tends to be higher, suggesting that there are relatively more employees earning middle-range wages.
- The average prevailing wage for both certified and denied visa applications is nearly the same.
- The year in which the employer company was established does not have a significant effect on whether the visa is certified or denied.
- 43.2% of visa applications are approved for applicants from Asia, the highest percentage among regions. However, the denial rate for visas is also elevated for Asia, at 22.9%.
- 66.7% of visa applications are approved. Among these approvals, 25% are awarded to employees with a bachelor's degree, making them the largest group, followed by those with a master's degree.
- 64.7% of employees with job experience are granted a visa.
- 88.2% of employees who do not require job training are granted a visa, a significantly higher percentage compared to those who do need job training.
- Employment opportunities in the Northeast (26.5%) and West regions (25%) of the U.S. have higher visa approval rates compared to the Island region, which has the lowest approval rates.
- 81% of visa denials occur in employment opportunities with annual wage structures.
- Both visa denials and approvals are prevalent for full-time job requirements, but the rate of approval is higher.

5. Data Preprocessing

5.1. Duplicate value check

- No duplicate values are found in the data set.

5.2. Missing value treatment

- No missing values or null values are found in the data set.

5.3. Outlier detection

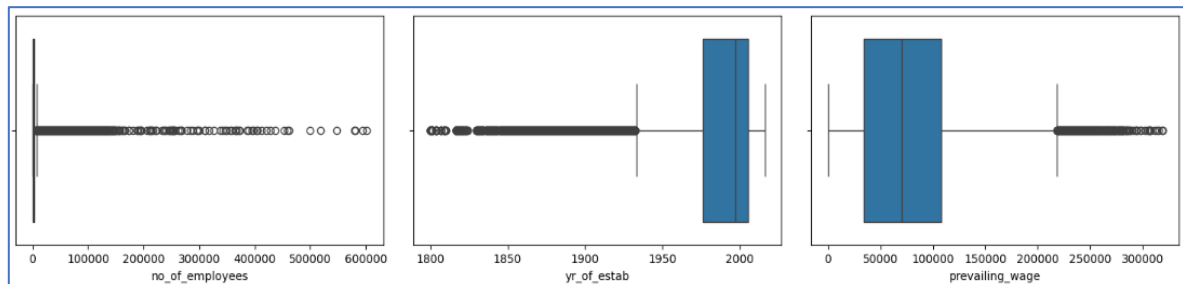


Fig 24. Outlier detection

OBSERVATIONS:

- There are outliers in the data.
- However, we will not treat them as they are proper values.

5.4. Feature Engineering

- The case_id column has unique values, which are not important to data set. Hence the column is dropped from the data set.
- The columns has_job_experience, requires_job_training and full_time_position comprise 'Yes' and 'No' entries, which are converted into binary 1 and 0 respectively.
- The categorical columns continent, education_of_employee, region_of_employment and unit_of_wage have been passed numerical values for imputation.

	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment
0	0	0	0	0	14513	2007	2
1	0	2	1	0	2412	2002	0
2	0	1	0	1	44444	2008	2
3	0	1	0	0	98	1897	2
4	1	2	1	0	1082	2005	1

prevailing_wage	unit_of_wage	full_time_position	case_status
592.203	0	1	Denied
83425.650	3	1	Certified
122996.860	3	1	Denied
83434.030	3	1	Denied
149907.390	3	1	Certified

Table 5. Numerical columns imputed data set

5.5. Data preparation

- The target variable 'case_status' is dropped from the data set.
- The data set has been split into training, validation and test set with a split of 60%, 20% and 20% respectively.

```
(15288, 10) (5096, 10) (5096, 10)

print("Number of rows in train data =", X_train.shape[0])
print("Number of rows in validation data =", X_val.shape[0])
print("Number of rows in test data =", X_test.shape[0])

Number of rows in train data = 15288
Number of rows in validation data = 5096
Number of rows in test data = 5096
```

Table 6. Train-Validation-Test Split

- Negative values are usually indicative of data errors or misinterpretations. In the present data set, negative values in the no_of_employees column is found to be an error. Dropping the rows with negative values, leads to loss of data. Hence the negative values in the column are converted to zero in the train, test and validation data set.
- The encoded variables are reverse mapped to convert them into categorical values.

Train data set

```
cols = X_train.select_dtypes(include=["object", "category"])
for i in cols.columns:
    print(X_train[i].value_counts())
    print("\n" * 30)
```

```
continent
Asia      10085
Europe    2285
North America  1944
South America  528
Africa     333
Oceania    113
Name: count, dtype: int64
*****

education_of_employee
Bachelor's  6141
Master's    5792
High School 2045
Doctorate   1310
Name: count, dtype: int64
*****

region_of_employment
Northeast  4312
South      4248
West       3920
Midwest    2576
Island     232
Name: count, dtype: int64
*****

unit_of_wage
Year      13786
Hour       1286
Week       156
Month       60
Name: count, dtype: int64
*****
```

Table 7. Reverse mapped variables in Train set

Validation data set

```
cols = X_val.select_dtypes(include=["object", "category"])
for i in cols.columns:
    print(X_val[i].value_counts())
    print("\n" * 30)
```

```
continent
Asia      3395
Europe    713
North America  655
South America  173
Africa     121
Oceania     39
Name: count, dtype: int64
*****

education_of_employee
Bachelor's  2033
Master's    1886
High School  694
Doctorate    483
Name: count, dtype: int64
*****

region_of_employment
Northeast  1430
South      1389
West       1352
Midwest     855
Island       70
Name: count, dtype: int64
*****

unit_of_wage
Year      4576
Hour       452
Week       57
Month       11
Name: count, dtype: int64
*****
```

Table 8. Reverse mapped variables in Validation set

Test data set

```
cols = X_test.select_dtypes(include=["object", "category"])
for i in cols.columns:
    print(X_test[i].value_counts())
    print("*" * 30)
```

```
continent
Asia          3381
Europe         734
North America  693
South America  151
Africa         97
Oceania        40
Name: count, dtype: int64
*****
education_of_employee
Bachelor's    2060
Master's      1956
High School   681
Doctorate     399
Name: count, dtype: int64
*****
region_of_employment
Northeast    1453
South        1380
West         1314
Midwest      876
Island        73
Name: count, dtype: int64
*****
unit_of_wage
Year         4600
Hour          419
Week          59
Month         18
Name: count, dtype: int64
*****
```

Table 9. Reverse mapped variables in Test set

- The categorical columns are encoded to dummy variables using one-hot encoding.

6. Model building

The model can make wrong predictions as:

1. Predicting an applicant will receive a visa (positive outcome) when, in reality, the applicant should not receive a visa.
2. Predicting an applicant will not receive a visa (negative outcome) when, in reality, the applicant should receive a visa.

Out of the two, predicting an applicant will not receive a visa (negative outcome) when, in reality, the applicant should receive a visa is the important outcome to be considered.

The OFLC wants Recall to be maximized, greater the Recall higher the chances of minimizing false negatives. Hence, the focus should be on increasing Recall or minimizing the false negatives or in other words identifying the true positives (i.e. Class 1) so that the OFLC can provide better opportunities to identify and attract the right talent to work in US.

6.1. Model Building-Original Data

Recall is the suitable metric used to evaluate the performance of classification models. It measures the ability of a model to identify all relevant instances within a dataset. Specifically, recall is defined as the ratio of true positive predictions to the sum of true positive and false negative predictions.

The original data is considered to build the following models:

- **Bagging:** BaggingClassifier is used with a base DecisionTreeClassifier. This combination helps to reduce variance and prevent overfitting by aggregating multiple decision trees.
- **Random Forest:** RandomForestClassifier is an ensemble method that creates multiple decision trees and merges them to improve classification accuracy and control overfitting.
- **GBM:** GradientBoostingClassifier builds an ensemble of weak learners sequentially, where each new learner corrects errors made by the previous ones.
- **AdaBoost:** AdaBoostClassifier combines multiple weak classifiers to form a strong classifier by focusing on errors made by previous classifiers.
- **Decision Tree:** DecisionTreeClassifier is a simple yet powerful model that splits the data into subsets based on feature values.
- **XGBoost:** XGBClassifier is an optimized version of gradient boosting that is highly efficient and performs well with large datasets.

Recall Scores:

```

Training Performance:

Bagging: 0.9893241919686582
Random forest: 1.0
GBM: 0.8806072477962782
Adaboost: 0.8877571008814887
dtree: 1.0
XGB: 0.9409402546523017

Validation Performance:

Bagging: 0.7823149236192715
Random forest: 0.8413631022326674
GBM: 0.8757344300822562
Adaboost: 0.8821974148061105
dtree: 0.7432432432432432
XGB: 0.8545828437132785

```

Table 10. Original data: Recall scores for Training and Validation Set

Training and Validation performance difference:

```

Training and Validation Performance Difference:

Bagging: Training Score: 0.9893, Validation Score: 0.7823, Difference: 0.2070
Random forest: Training Score: 1.0000, Validation Score: 0.8414, Difference: 0.1586
GBM: Training Score: 0.8806, Validation Score: 0.8757, Difference: 0.0049
Adaboost: Training Score: 0.8878, Validation Score: 0.8822, Difference: 0.0056
dtree: Training Score: 1.0000, Validation Score: 0.7432, Difference: 0.2568
XGB: Training Score: 0.9409, Validation Score: 0.8546, Difference: 0.0864

```

Table 11. Original data: Training and Validation performance difference

OBSERVATIONS:

- The **GBM model (Validation Recall=0.875, Training and Validation performance difference=0.0049)** has the best performance followed by the **AdaBoost model (Validation Recall=0.882, Training and Validation performance difference=0.0056)** according to the validation results.
- It suggests that these models are effectively capturing the patterns in the data.

6.2. Model Building-Oversampled Data

- Oversampling is a technique used to address class imbalance in machine learning datasets. In scenarios where one class is significantly underrepresented compared to others, oversampling helps balance the dataset by increasing the number of instances of the minority class.
- **Synthetic Minority Over-sampling Technique (SMOTE)** is used for oversampling the data set. It generates synthetic samples of the minority class by interpolating between existing samples. It adds diversity to the minority class.

Counts of Labels and Shape of Train

```
Before Oversampling, counts of label 'Yes': 10210
Before Oversampling, counts of label 'No': 5078

After Oversampling, counts of label 'Yes': 10210
After Oversampling, counts of label 'No': 10210

After Oversampling, the shape of train_X: (20420, 21)
After Oversampling, the shape of train_y: (20420,)
```

Table 12. Oversampled Data: Counts of Labels and Shape of Train

Recall Scores:

```
Training Performance:

Bagging: 0.9815866797257591
Random forest: 1.0
GBM: 0.8431929480901077
Adaboost: 0.8272282076395691
dtree: 1.0
XGB: 0.9158667972575906

Validation Performance:

Bagging: 0.7511750881316098
Random forest: 0.8078730904817861
GBM: 0.831081081081081
Adaboost: 0.8152173913043478
dtree: 0.7253231492361927
XGB: 0.8437132784958872
```

Table 13. Oversampled data: Recall scores for Training and Validation Set

Training and Validation performance difference:

```
Training and Validation Performance Difference:

Bagging: Training Score: 0.9816, Validation Score: 0.7512, Difference: 0.2304
Random forest: Training Score: 1.0000, Validation Score: 0.8079, Difference: 0.1921
GBM: Training Score: 0.8432, Validation Score: 0.8311, Difference: 0.0121
Adaboost: Training Score: 0.8272, Validation Score: 0.8152, Difference: 0.0120
dtree: Training Score: 1.0000, Validation Score: 0.7253, Difference: 0.2747
XGB: Training Score: 0.9159, Validation Score: 0.8437, Difference: 0.0722
```

Table 14. Oversampled data: Training and Validation performance difference

OBSERVATIONS:

- The **AdaBoost model (Validation Recall=0.815, Training and Validation performance difference=0.0120)** has the best performance followed by the **GBM model (Validation Recall=0.831, Training and Validation performance difference=0.0121)** according to the validation results.
- It suggests that these models are effectively capturing the patterns in the data.

6.3. Model Building-Undersampled Data

- Undersampling is a technique used to address class imbalance by reducing the number of instances in the majority class. This can help to balance the class distribution and prevent the model from being biased towards the majority class.
- RandomUnderSampler** is used for undersampling the data set. It is a technique provided by the imblearn library to address class imbalance by reducing the number of instances in the majority class. By balancing the dataset, it helps improve model performance and mitigate biases towards the majority class.

Counts of Labels and Shape of Train

```
Before Under Sampling, counts of label 'Yes': 10210
Before Under Sampling, counts of label 'No': 5078

After Under Sampling, counts of label 'Yes': 5078
After Under Sampling, counts of label 'No': 5078

After Under Sampling, the shape of train_X: (10156, 21)
After Under Sampling, the shape of train_y: (10156,)
```

Table 15. Undersampled Data: Counts of Labels and Shape of Train

Recall Scores:

```
Training Performance:

Bagging: 0.9679007483261126
Random forest: 1.0
GBM: 0.7473414730208744
Adaboost: 0.7172115005907838
dtree: 1.0
XGB: 0.8812524615990548

Validation Performance:

Bagging: 0.6125146886016452
Random forest: 0.6797884841363102
GBM: 0.7282608695652174
Adaboost: 0.7074030552291422
dtree: 0.6354289071680376
XGB: 0.6891891891891891
```

Table 16. Undersampled data: Recall scores for Training and Validation Set

Training and Validation performance difference:

```
Training and Validation Performance Difference:

Bagging: Training Score: 0.9679, Validation Score: 0.6125, Difference: 0.3554
Random forest: Training Score: 1.0000, Validation Score: 0.6798, Difference: 0.3202
GBM: Training Score: 0.7473, Validation Score: 0.7283, Difference: 0.0191
Adaboost: Training Score: 0.7172, Validation Score: 0.7074, Difference: 0.0098
dtree: Training Score: 1.0000, Validation Score: 0.6354, Difference: 0.3646
XGB: Training Score: 0.8813, Validation Score: 0.6892, Difference: 0.1921
```

Table 17. Undersampled data: Training and Validation performance difference

OBSERVATIONS:

- The **AdaBoost model (Validation Recall=0.707, Training and Validation performance difference=0.0098)** has the best performance followed by the **GBM model (Validation Recall=0.728, Training and Validation performance difference=0.019)** according to the validation results.
- It suggests that these models are effectively capturing the patterns in the data.

7. Model Performance Improvement using Hyperparameter Tuning

- After building 18 models from **Bagging, Random forest, GBM, Adaboost, dtree and XGB**, it was observed that both the **GBM (Model 1) and Adaboost models (Model 2), trained on original dataset**, as well as the **Adaboost model (Model 3) trained on an undersampled dataset**, exhibited strong performance on both the training and validation datasets.
- Sometimes models might overfit after undersampling and oversampling, so it's better to tune the models to get a generalized performance.
- Hyperparameter tuning is done for these 3 models using the same data as it was trained on before.
- **RandomizedSearchCV** is used for tuning. It is a technique provided by scikit-learn for hyperparameter optimization. It allows to efficiently search through a subset of hyperparameter values to find the best model configuration. It randomly samples from a defined range of hyperparameter values, which can be more efficient and computationally feasible.

7.1. Tuning GBM model with Original data

CV score:

Best parameters are {'subsample': 0.5, 'n_estimators': 125, 'max_features': 1, 'learning_rate': 0.01, 'init': AdaBoostClassifier(random_state=1)} with **CV score=0.9878550440744368**:

CPU times: total: 3.53 s

Wall time: 1min 22s

Parameters:

```
tuned_gbm1 = GradientBoostingClassifier(
    random_state=1,
    subsample=0.9,
    n_estimators=150,
    max_features=0.5,
    learning_rate=0.01,
    init=AdaBoostClassifier(random_state=1),
)
tuned_gbm1.fit(X_train, y_train)
```

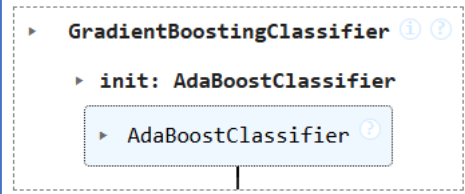


Table 18. GBM-Original Parameters

Model's performance on training set

	Accuracy	Recall	Precision	F1
0	0.737	0.915	0.748	0.823

Table 19. GBM (Original) model performance-Train set

Model's performance on validation set

	Accuracy	Recall	Precision	F1
0	0.740	0.912	0.752	0.824

Table 20. GBM (Original) model performance-Validation set

7.2. Tuning Adaboost model with original data

CV score:

Best parameters are {'n_estimators': 30, 'learning_rate': 0.05, 'estimator': DecisionTreeClassifier(max_depth=1, random_state=1)} with **CV score=0.9729676787463271**:
 CPU times: total: 2.23 s
 Wall time: 50.3 s

Parameters:

```
tuned_adb = AdaBoostClassifier(
    random_state=1,
    n_estimators=20,
    learning_rate=0.1,
    estimator=DecisionTreeClassifier(max_depth=2, random_state=1),
)
tuned_adb.fit(X_train_un, y_train_un)
```

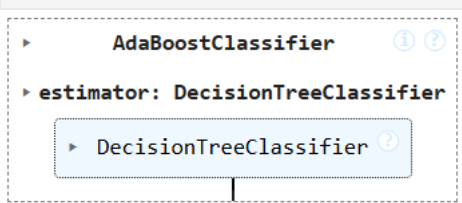


Table 21. Adaboost- Original Parameters

Model's performance on training set

	Accuracy	Recall	Precision	F1
0	0.733	0.909	0.747	0.820

Table 22. Adaboost (Original) model performance-Train set

Model's performance on validation set

	Accuracy	Recall	Precision	F1
0	0.737	0.909	0.751	0.822

Table 23. Adaboost (Original) model performance- Validation set

7.3. Tuning Adaboost model with Undersampled data

CV score:

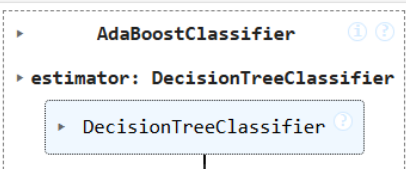
Best parameters are {'n_estimators': 10, 'learning_rate': 0.01, 'estimator': DecisionTreeClassifier(max_depth=1, random_state=1)} with **CV score=0.9312726426438074**:

CPU times: total: 1.09 s

Wall time: 14.6 s

Parameters:

```
tuned_adb1 = AdaBoostClassifier(
    random_state=1,
    n_estimators=20,
    learning_rate=0.1,
    estimator=DecisionTreeClassifier(max_depth=2, random_state=1),
)
tuned_adb1.fit(X_train_un, y_train_un)
```



The image shows a Jupyter Notebook parameter tree for an AdaBoostClassifier. The root node is 'AdaBoostClassifier' with a question mark icon. It has a child node 'estimator: DecisionTreeClassifier'. This node has a further child node 'DecisionTreeClassifier' with a question mark icon.

Table 24. Adaboost-Undersampled Parameters

Model's performance on training set

	Accuracy	Recall	Precision	F1
0	0.702	0.721	0.694	0.707

Table 25. Adaboost (Undersampled) model performance-Train set

Model's performance on validation set

	Accuracy	Recall	Precision	F1
0	0.710	0.717	0.826	0.767

Table 26. Adaboost (Undersampled) model performance-Validation set

7.4. Insights on Model Tuning

- **GBM (Model 1)** and **Adaboost models (Model 2)**, trained on original dataset and **Adaboost model (Model 3)** trained on an undersampled dataset exhibit a CV score of 0.987, 0.972 and 0.9312.
- The score of approximately 0.987 suggests that the model with the given hyperparameters achieves an average performance of about 98.7% on the cross-validation folds.
- This is a high score and indicates that the model performs well on unseen data, given the selected hyperparameters.
- Out of the three models hypertuned, the **Model 1 and Model 2** show high recall of **0.912 and 0.909** respectively on validation data sets.
- The **F1 scores** of the above two models are also close to 1, **0.824 and 0.822** respectively.
- The difference between the train and validation set is seen to be 0.003, 0 and 0.004 for Model 1, Model 2 and Model 3 respectively.

8. Model Performance Comparison and Final Model Selection

The accuracy, recall, precision and F1 metrics for Model 1, Model 2 and Model 3 of training and validation sets are obtained.

Training performance comparison:			
	Gradient boosting trained with Original data	AdaBoost trained with Original data	AdaBoost trained with Undersampled data
Accuracy	0.737	0.733	0.702
Recall	0.915	0.909	0.721
Precision	0.748	0.747	0.694
F1	0.823	0.820	0.707

Table 27. Training performance comparison

Validation performance comparison:			
	Gradient boosting trained with Original data	AdaBoost trained with Original data	AdaBoost trained with Undersampled data
Accuracy	0.740	0.737	0.710
Recall	0.912	0.909	0.717
Precision	0.752	0.751	0.826
F1	0.824	0.822	0.767

Table 28. Validation performance comparison

OBSERVATIONS:

- **Model 1** and **Model 2** have similar recall scores (**0.912** and **0.909**, respectively), which are higher than Model 3's recall (0.717). As minimizing false negatives is crucial, Models 1 and 2 are preferable due to their higher recall.
- The difference between trained and validation data set recall values are low for both Model 1 (0.003) and Model 2 (0) trained under original data set.
- Both the models seem to be performing well. The recall score suggests good generalization and consistent performance.
- Model 1 has the highest F1 score (0.824), closely followed by Model 2 (0.822). F1 score balances precision and recall, so considering an overall balance between the two metrics, Models 1 and 2 are preferable.
- In order to further select the model, the metrics is measured for test data for Model 1 and Model 2.

The accuracy, recall, precision and F1 metrics for Model 1 and Model 2 of test sets are obtained.

GBM-Original Test data

	Accuracy	Recall	Precision	F1
0	0.729	0.913	0.742	0.818

Table 29. GBM (Original) model performance-Test set

Adaboost-Original Test data

	Accuracy	Recall	Precision	F1
0	0.725	0.906	0.741	0.815

Table 30. Adaboost (Original) model performance-Test set

OBSERVATIONS:

- Based on the test data sets, **Model 1 (GBM trained with original data set)** has high recall (0.913) and F1 (0.818). Hence Model 1 can be considered as the primary model, followed by **Model 2 (Adaboost trained with original data set)**.
- This performance is in line with what is achieved with this model on the train and validation sets.
- Thus, **Model 1 is the most generalized model**, with Model 2 following closely.

9. Feature importance

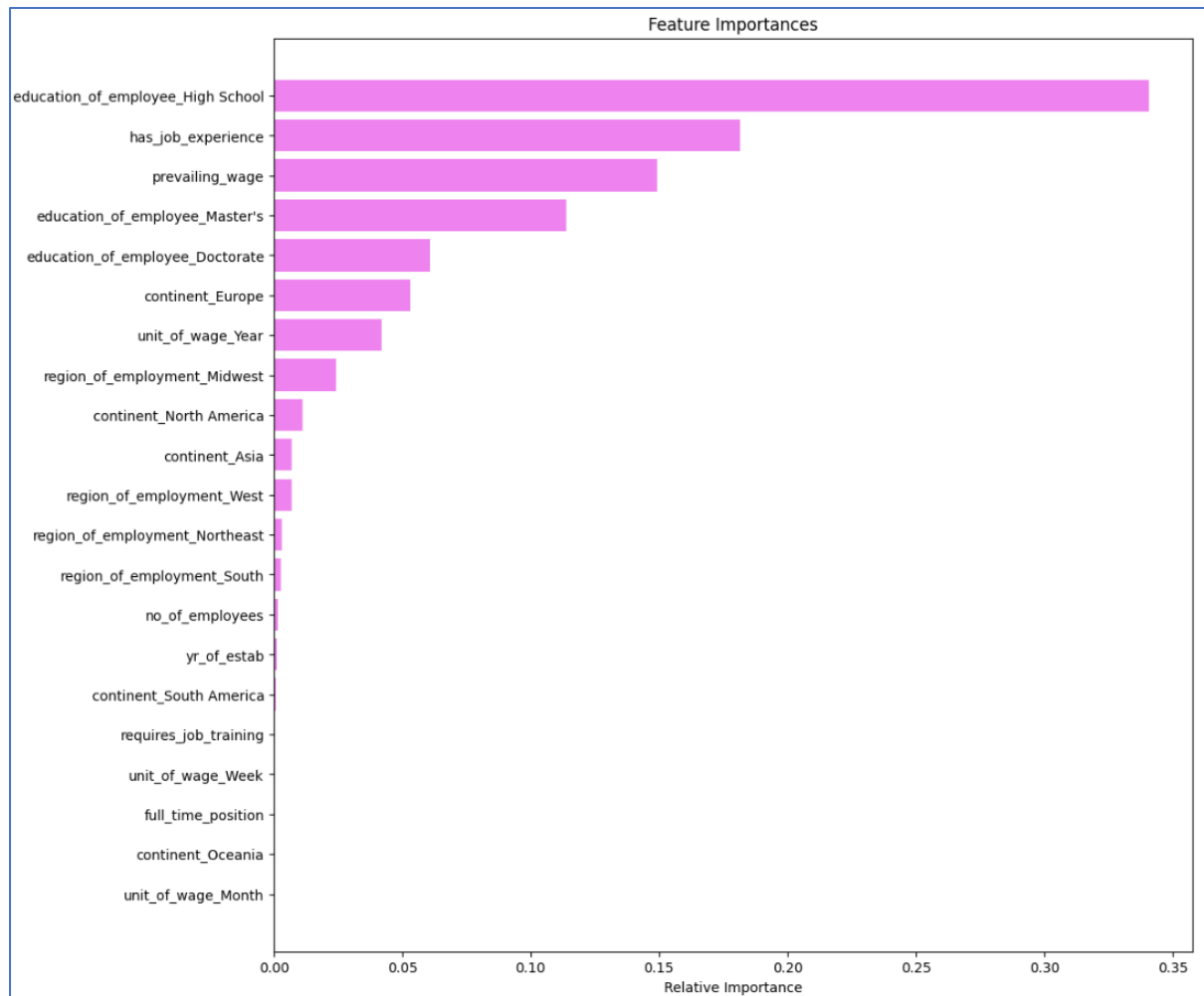


Fig 25. Feature importance

OBSERVATION:

- The **education_of_employee High School**, **has_job_experience** and **prevailing_wage** are the important features for making predications on the visa approval process.

10. Actionable Insights & Recommendations

- A predictive model has been developed to assess employees who are likely to be approved for visas. By identifying the key factors driving the visa approval process, the OFLC can take appropriate actions to select the right candidates for employment in the U.S.
- Factors that drive visa approval process includes education of employee (High school), has the employee have a job experience and prevailing wages provided by the company.
- Employees with only a high school education are typically denied visa approval, as a bachelor's degree is generally required for jobs in the U.S. Most companies necessitate at least a graduate-level education. While a master's or doctorate is not mandatory, having a basic level of education in the relevant field is essential.
- Nearly 65% of employees with prior job experience are granted visas, indicating that having previous work experience is often essential for companies when offering job opportunities. The time and resources invested in training new hires are significant considerations for companies. Therefore, it is crucial to evaluate an immigrant's prior experience when considering their visa approval process.
- Prevailing wage is also a crucial factor in visa approval, as the investment in an employee's compensation is a key consideration for hiring. Ensuring that the offered wage meets or exceeds industry standards is essential for securing approval.
- The number of employees, the year a company was established, or whether a job is full-time are not significant factors in the visa approval or denial process.